

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2026

SA1-Health and Care

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) You are the lead health actuary at a consulting firm. Your client is a large domestic diversified conglomerate that currently owns a moderately successful domestic life insurance company.

Recently, the Government increased the Foreign Direct Investment (FDI) limit in the insurance sector to 100%. Following this, a major foreign multinational health insurer, approached your client to explore a partnership to enter the local health insurance market.

The Board of the conglomerate is evaluating whether to launch a new Stand-Alone Health Insurance (SAHI) Joint Venture with foreign multinational health insurer, or to simply inject capital into existing life insurance company to expand its existing, portfolio of health riders.

- i) Critically evaluate the macro-economic, competitive, and regulatory implications of the government's decision to allow 100% FDI in the health insurance sector. Discuss how this policy shift alters the strategic landscape for domestic incumbents, healthcare providers, and policyholders. **[10]**
- ii) Prepare a briefing note for the Board comparing the strategic, operational, regulatory, and capital implications of launching a new SAHI Joint Venture versus expanding the health insurance operations within the existing Life insurance Company. **[14]**

The Board has decided to proceed with establishing the SAHI Joint Venture. To differentiate from local competitors who focus heavily on basic hospitalisation (indemnity) covers, the CEO wants the SAHI's flagship launch products to be comprehensive Individual Critical Illness (CI) and long-term Individual Income Protection (IP). Foreign multinational health insurer wishes to import its product designs and pricing methodology for these products.

- iii) Discuss the actuarial, commercial, and risk management challenges of pricing and reserving for these specific products in the local market. Recommend strategies the new SAHI could adopt to mitigate these risks, particularly regarding data scarcity, product design, and reinsurance. **[18]**

Instead of an organic launch, an opportunity has arisen for the Joint Venture to acquire a mid-sized, struggling domestic Stand-Alone Health Insurance (SAHI) company. SAHI company's portfolio consist of:

- 85% Individual and Family Floater Medical Indemnity: Currently experiencing a loss ratio of 115% due to severe medical inflation and poor provider network management.
- 15% Individual CI and IP: Highly profitable, with excellent persistency.

- iv) Discuss the specific actuarial investigations and due diligence you would undertake to assess the viability of acquiring SAHI. Your answer should highlight how the distressed Medical Indemnity portfolio impacts the valuation of the target, the assessment of its actuarial liabilities, and the post-acquisition integration strategy. **[18]**

[60 marks]

Q.2) You are the Chief Actuary of ABC Health, a mid-sized Standalone Health Insurance company. As part of the expansion, a large multinational manufacturing firm, XYZ Corp, has approached ABC Health. XYZ Corp. currently operates an internally funded (uninsured) Post-Retirement Medical Benefit (PRMB) scheme for its former employees.

XYZ Corp is looking to completely offload this risk by purchasing a single-premium, fully insured group policy from ABC Health to cover all future medical claims of its current retirees.

- i) Discuss the commercial rationale for XYZ Corp wanting to move this scheme to a fully insured arrangement. Outline the potential disadvantages to this decision for XYZ Corp. [5]
- ii) Detail and discuss the specific data, historical information, and management information you would request from XYZ Corp to assess and price this liability. [10]
- iii) Describe the methodology and actuarial assumptions you would employ to price this single-premium PRMB buy-out policy. You should specifically address how you will allow for medical inflation, data credibility, expenses, and the long-term nature of the risk. [5]

Currently, ABC Health's balance sheet consists solely of short-tail retail indemnity business. Taking on the XYZ Corp PRMB buy-out will introduce a large block of long-term liabilities to the company.

- iv) Outline the major asset and liability classes currently likely to be held by ABC Health prior to this transaction. [10]
 - v) Assess how acquiring the PRMB buy-out portfolio will impact ABC Health's Asset Liability Management (ALM) and capital strategy. You should highlight the practical challenges of duration matching and inflation hedging for long-term healthcare liabilities in the current economic environment. [10]
- [40 marks]**
